A twelve-variable Diophantine candidate

Strictly positive integer unknowns; 95 evaluator operations.

The only unknowns are $a, b, c, d, e, f, g, h, i, j, k, \ell \in \mathbb{Z}_{>0}$. Each $\texttt{r01}, \ldots, \texttt{r95}$ below is an abbreviation, not an additional unknown. Evaluate the rows in order. The equation is $\boxed{\texttt{r95} = 0}$.

$\texttt{r01} = a \cdot \beta$
$\texttt{r02} = 1 + \texttt{r01}$
$\texttt{r03} = \texttt{r02} \cdot \texttt{r02}$
$\texttt{r04} = \texttt{r02} \cdot \texttt{r03}$
$\texttt{r05} = g \cdot \texttt{r03}$
$\texttt{r06} = \texttt{r02} - b$
$\texttt{r07} = \texttt{r06} - c$
$\texttt{r08} = f \cdot g$
$\texttt{r09} = i + j$
$\texttt{r10} = g \cdot \texttt{r09}$
$\texttt{r11} = i + \texttt{r10}$
$\texttt{r12} = \texttt{r11} \cdot \texttt{r11}$
$\texttt{r13} = \texttt{r08} + g$
$\texttt{r14} = 4 \cdot \texttt{r13}$
$\texttt{r15} = 3 + \texttt{r14}$
$\texttt{r16} = \texttt{r13} \cdot \texttt{r13}$
$\texttt{r17} = \texttt{r15} + \texttt{r16}$
$\texttt{r18} = \texttt{r13} \cdot \texttt{r11}$
$\texttt{r19} = f + \texttt{r18}$
$\texttt{r20} = \texttt{r08} \cdot \texttt{r10}$
$\texttt{r21} = \texttt{r04} \cdot \texttt{r01}$
$\texttt{r22} = \kappa \cdot \texttt{r02}$
$\texttt{r23} = f + \texttt{r22}$
$\texttt{r24} = \texttt{r07} \cdot \texttt{r23}$
$\texttt{r25} = b \cdot \texttt{r02}$
$\texttt{r26} = \texttt{r24} - \texttt{r25}$
$\texttt{r27} = \texttt{r20} + \texttt{r09}$
$\texttt{r28} = \texttt{r27} + h$
$\texttt{r29} = \texttt{r09} \cdot \texttt{r26}$
$\texttt{r30} = \texttt{r28} - \texttt{r29}$
$\texttt{r31} = \texttt{r21} \cdot \texttt{r30}$
$\texttt{r32} = g \cdot \texttt{r01}$
$\texttt{r33} = \texttt{r31} + \texttt{r32}$
$\texttt{r34} = \texttt{r04} \cdot \texttt{r26}$
$\texttt{r35} = \texttt{r33} - \texttt{r34}$
$\texttt{r36} = \texttt{r35} \cdot \texttt{r35}$
$\texttt{r37} = \texttt{r21} \cdot \texttt{r21}$
$\texttt{r38} = \texttt{r20} \cdot \texttt{r27}$
$\texttt{r39} = 1 + \texttt{r38}$
$\texttt{r40} = \texttt{r37} \cdot \texttt{r39}$
$\texttt{r41} = \texttt{r36} - \texttt{r40}$
$\texttt{r42} = \texttt{r06} \cdot \texttt{r01}$
$\texttt{r43} = c + \texttt{r42}$
$\texttt{r44} = \texttt{r03} - 1$
$\texttt{r45} = \texttt{r43} \cdot \texttt{r44}$
$\texttt{r46} = \nu \cdot \texttt{r02}$
$\texttt{r47} = \mu + \texttt{r46}$
$\texttt{r48} = a \cdot \texttt{r47}$
$\texttt{r49} = \texttt{r45} + \texttt{r48}$
$\texttt{r50} = \texttt{r09} - \texttt{r49}$
$\texttt{r51} = \texttt{r50} - 1$
$\texttt{r52} = \texttt{r05} \cdot \ell$
$\texttt{r53} = \texttt{r08} \cdot \texttt{r02}$
$\texttt{r54} = \texttt{r52} + \texttt{r53}$
$\texttt{r55} = \texttt{r54} - \texttt{r51}$
$\texttt{r56} = \texttt{r05} \cdot \texttt{r19}$
$\texttt{r57} = \texttt{r15} \cdot \texttt{r55}$
$\texttt{r58} = \texttt{r56} + \texttt{r57}$
$\texttt{r59} = \texttt{r58} \cdot \texttt{r58}$
$\texttt{r60} = \texttt{r05} \cdot \texttt{r05}$
$\texttt{r61} = \texttt{r17} \cdot \texttt{r12}$
$\texttt{r62} = 1 + \texttt{r61}$
$\texttt{r63} = \texttt{r60} \cdot \texttt{r62}$
$\texttt{r64} = \texttt{r59} - \texttt{r63}$
$\texttt{r65} = 1 + d$
$\texttt{r66} = \texttt{r65} \cdot \texttt{r65}$
$\texttt{r67} = \texttt{r66} - 1$
$\texttt{r68} = \texttt{r17} \cdot \texttt{r67}$
$\texttt{r69} = 1 - \texttt{r68}$
$\texttt{r70} = \texttt{r12} \cdot \texttt{r12}$
$\texttt{r71} = e \cdot \texttt{r11}$
$\texttt{r72} = \texttt{r71} - \texttt{r49}$
$\texttt{r73} = \texttt{r66} \cdot \texttt{r72}$
$\texttt{r74} = \texttt{r73} - \texttt{r11}$
$\texttt{r75} = \texttt{r74} + k$
$\texttt{r76} = \texttt{r75} \cdot \texttt{r75}$
$\texttt{r77} = \texttt{r74} \cdot \texttt{r74}$
$\texttt{r78} = \texttt{r68} \cdot \texttt{r77}$
$\texttt{r79} = \texttt{r78} - 1$
$\texttt{r80} = \texttt{r69} \cdot \texttt{r76}$
$\texttt{r81} = \texttt{r80} + \texttt{r79}$
$\texttt{r82} = \texttt{r68} \cdot \texttt{r69}$
$\texttt{r83} = \texttt{r70} \cdot \texttt{r81}$
$\texttt{r84} = \texttt{r83} - \texttt{r82}$
$\texttt{r85} = \texttt{r84} \cdot \texttt{r84}$
$\texttt{r86} = \texttt{r70} \cdot \texttt{r82}$
$\texttt{r87} = 4 \cdot \texttt{r86}$
$\texttt{r88} = \texttt{r87} \cdot \texttt{r79}$
$\texttt{r89} = \texttt{r85} + \texttt{r88}$
$\texttt{r90} = 1 + \texttt{r41}$
$\texttt{r91} = 1 + \texttt{r64}$
$\texttt{r92} = \texttt{r90} \cdot \texttt{r91}$
$\texttt{r93} = 1 + \texttt{r89}$
$\texttt{r94} = \texttt{r92} \cdot \texttt{r93}$
$\texttt{r95} = \texttt{r94} - 1$

The four fixed integers. All indices in the following formulas are locally bound indices for constants, not unknowns. Set $\rho = 2^{67}$ and $\beta = \rho^{8807} - 1$. Let the fixed set and ordered list be

$$\mathcal{S} = \big(\{0,\dots,942\} \cup \{983,\dots,992\} \cup \{1003,\dots,1011\} \cup \{1013,\dots,1031\} \\ \cup \{1063, 1064, 1095, 2195, 3296\}\big) \setminus \{1\},$$
$$\lambda = (1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, 1015, 1016, 1017, 1018, 1019, 1020, 1021, 1003, 1004, 1005, 1006, 1007, 1008,$$

The list λ has 41 entries, numbered from zero. The fixed list σ has 943 entries, also numbered from zero, and is printed in decimal on the following pages. Then

$$\mu = \beta - \sum_{u\in\mathcal{S}} \rho^u - 2\rho^{1063},$$
$$\nu = \beta + 4 + \sum_{u=1013}^{1022} \rho^u + \sum_{u=1065}^{1073} \rho^u + \rho^{2195} + \rho^{3296} \\ + (2^{51}-2)(\rho^{4393} + \rho^{5495}) + (2^{65}-1) \sum_{u=8792}^{8804} \rho^u + (2^{19}-1)\rho^{8805},$$
$$\kappa = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ + \sum_{u=0}^{942} \sigma_u(\rho^{4393-u} + \rho^{5495-u}) + \sum_{u=0}^{40} 2^{10+u}(\rho^{4393-\lambda_u} + \rho^{5495-\lambda_u}).$$

Decimal selector list. Read from left to right, top to bottom; each 14-digit block is one decimal integer. Leading zeroes do not change its value. There are exactly 943 blocks.

17557309618177 17592186039297 17592186044417 02542620639233 17427232456705 17358378762241 17289525067777
17220671373313 17151817678849 17014244507649 16876671336449 16807817641985 16738963947521 16463951822849
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